Project Title: Subtitle Describing the Main Objective

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Abstract

This project investigates [insert core problem/dataset] using exploratory data analysis and statistical modeling techniques. Using data sourced from [insert source], we apply [insert key methods] to evaluate [insert main hypothesis/research question]. Our primary findings demonstrate that [insert key quantitative result]. These results suggest [insert broader implication for data science/domain].

Contents

1 Introduction **1**

2 Smart Location Database **2**

3 Methods **3**

3.1 Method Subsection 3

4 Results **4**

5 Conclusion **4**

6 Future Work **4**

References **4**

1 Introduction

For most American households, car ownership is often the single largest expense after housing. It is a financial reality that can become severe when urban planning mandates vehicle ownership. In transit-poor and sprawling neighborhoods, owning multiple automobiles is rarely a luxury, but instead a compulsory toll required to access basic employment, education, and healthcare. This forced reliance acts as a pervasive "car tax," disproportionately impacting low-income families through auto loans, insurance, and maintenance simply to participate in the regional economy. When infrastructure fails to provide viable transportation alternatives, vehicle ownership ceases to represent personal freedom and instead becomes an enforced mechanism of inequality.

Urban equity is the idea that all people, regardless of income, race, or geographic location, deserve fair and affordable access to the opportunities and public infrastructure that enable socio-economic mobility. In transportation planning, urban equity is crucial because mobility directly dictates life outcomes. Without reliable transportation to employment centers, quality healthcare, fresh food, and education, marginalized communities face significant barriers to escaping poverty.

A central driver of urban inequity is the transit gap: the disparity between where public transit dependent populations live and where high-quality and frequent public transportation service is actually provided. Bridging the transit gap is important because when public transit options fall short of community needs, low-income households are either forced into high-cost vehicle ownership or left isolated from economic

opportunities. Closing this gap ensures that low-income residents can reliably reach jobs and essential services.

The transit gap has a long research history. It began in 1968, when John F. Kain proposed the Spatial Mismatch Hypothesis, which describes a physical separation between where low-income and minority workers live and where entry-level jobs are located. (Kain, 1968) His study focuses on how residential segregation combined with the suburbanization of entry-level jobs creates a spatial disconnect where physical distance and inadequate transit drive systemic unemployment.

Thirty years later, Litman focuses on the difference between horizontal equity (fair allocation of resource) and vertical equity (allocating resources to compensate for socio-economic disadvantage). (Litman, 2002) The work demonstrates how failing to the close the transit gap exacerbates income equality and social isolation. Grengs analyzes the transit gap in Detroit where he demonstrates that public transit riders face severe access penalites compared to car owners. (Grengs, 2010) He argues that expanding transit access is far more effective for urban equity than continuing to subsidize auto-oriented development.

In 2013, Jiao and Dillivan coin the term "Transit Desert". (Jiao & Dillivan, 2013) They formulate a spatial methodology comparing transit-dependent populations (demand) with public transit infrastructure (supply) across urban neighborhoods. The work shows that the areas with the highest transit needs often receive disproportionately low transit investment, exacerbating urban inequality. Blumenberg and Pierce focus on the intersection of household poverty, forced vehicle ownership, and transit dependence. (Blumenberg & Pierce, 2014) They show that while automobiles significantly expand job access, the financial strain of maintaining a car in transit-poor areas often offsets economic gains, reinforcing the need for reliable public transit alternatives.

The Omaha-Council Bluffs metropolitan region has historically evolved around low-density and car-centric suburban growth, requiring most households to own at least one vehicle. The decline of early streetcar networks coupled with highway expansion encouraged suburbanization, leaving local transportation infrastructure structured for private automobile ownership. Consequently, the transit gap presents a major hurdle for urban equity. With major commercial corridors, healthcare facilities, and industrial employment hubs being widely spread, a severe transit gap places a disproportionate financial burden on low-income residents, isolates non-driving populations, and limits regional socio-economic mobility.

This study investigates the transit gap in the Omaha-Council Bluff metropolitan area through four distinct analytical lenses: 1) Clustering neighborhoods into archetypes; 2) Determining drivers of auto dependence through prediction models; 3) ; 4) . The intention is to better understand the transit gap, identify the areas it impacts the most, and make strategic policy recommendations for city planners and regional managers such as the Metropolitan Area Planning Agency (MAPA) and Metro Transit.

The paper is divided into the following sections: a description of the Smart Location Database used for analysis; Methods defining experimental and statistical models; Results stating the findings, their significance, and policy recommendations; a Conclusion; and Future Work detailing how to further discovery and tools creation.

2 Smart Location Database

The EPA Smart Location Database (SLD) is a nationwide and open-source geographic resource developed by the U.S. Environmental Protection Agency. (U.S. Environmental Protection Agency, 2021) It is designed to measure and analyze built environment variables across census block groups. Designed for urban planners, transportation modelers, and policy researchers, the SLD provides standardized indicators structured around the "Five D's" of urban planning: Density, Diversity, Design, Destination Accessibility, and Distance to Transit. By aggregating different spatial metrics including gross residential density, street intersection frequency, job-housing mix, transit service availability, and regional job accessibility into a single dataset, the database allows analysts to evaluate how neighborhood structure shapes regional travel behavior.

The primary aim of the Smart Location Database is to help create sustainable, resilient, and transit-supported communities by providing an empirical foundation for land-use and transportation decisions. By offering

accessible and consistent spatial indicators across all U.S. metropolitan regions, the database empowers researchers to predict travel demand, evaluate vehicle miles traveled, model household auto ownership, and identify areas with pronounced walkability or transit accessibility gaps. Ultimately, the SLD seeks to support multi-modal planning initiatives that lower greenhouse gas emissions, enhance urban mobility options, and inform equitable land development strategies across diverse regional areas.

The SLD is a vast dataset consisting of 220,740 rows with a size of almost 200 MB. Since the Omaha-Council Bluffs metropolitan area resides in Douglas, Sarpy, and Pottawattamie counties, the tigris R-package was utilized to identify the specific census blocks of the region. The rows matching these blocks were extracted and placed in their own file with a final 728 rows and size of roughly 663 KB. This smaller, focused dataset allowed for more efficient analysis of the metropolitan area.

It should be noted that the SLD makes use of sentinel values for missing data. For example, the Distance to Transit Stop feature uses the value "-99999" for census blocks that have no regional transit. These values are handled by omission or recoding based on the analysis utilized as clearly stated in the Methods section.

The SLD is a rich dataset of 90 unique features. We focus on a subset that is relevant to our analysis of urban equity in the Omaha-Council Bluffs Metro area. The utilized variables are:

- **Census Block Group Identifier (GEOID):** A unique 12-digit Federal Information Processing Series (FIPS) code that uniquely identifies each Census Block Group nationwide.
- **Distance to Nearest Transit Stop (D4A):** Distance (in meters) from the block group population centroid to the nearest transit stop.
- **Employment & Housing Mix (D2B_E8MIX):** Balance between housing units and employment opportunities within the block group. Scale: 0 = Pure residential or pure commercial; 1 = Highly balanced mix of jobs and housing.
- **Gross Employment Density (D1C):** Total jobs per acre of unprotected land. Higher values indicate higher concentrations of employment.
- **Gross Residential Density (D1A):** Housing units per acre of unprotected land. Higher values indicate denser residential settlement.
- **Median Household Income (MedIncome):** Census block group median household income in USD.
- **National Walkability Index (NatWalkInd):** Composite index evaluating overall pedestrian friendliness based on density, diversity, and design. Scale: 1 = Least Walkable (Auto-Dependent Car Suburb); 20 = Most Walkable (Pedestrian-Oriented Urban Core).
- **Percentage Low-Income Households (Pct_LowInc):** Proportion of households living below the regional low-income threshold.
- **Street Intersection Density (D3B):** Pedestrian-oriented street grid density (intersections with 3 or more legs per square mile). Scale: Higher values reflect a tighter, more interconnected street grid conducive to walking.

By focusing on the five D's (Density, Diversity, Design, Destination Accessibility, and Distance to Transit), these features allow our study to analyze the presence and impact of transit gaps in the Omaha-Council Bluffs metropolitan area. They provide a comprehensive view of the built environment, socio-economic conditions, and accessibility factors that influence urban equity in the region. The variables permit us to determine unique neighborhood archetypes, predict the primary influences of compulsory automobile ownership, and...

3 Methods

General description

3.1 Method Subsection

Content

4 Results

Content

5 Conclusion

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6 Future Work

"Be very specific" Include both how and why

References

Blumenberg, E., & Pierce, G. (2014). Multimodal travel and poverty: With and without an auto. *Urban Studies*, *51*(1), 58–74. https://doi.org/10.1177/0042098013484532

Grengs, J. (2010). Job accessibility and the modal divide in detroit. *Journal of the American Planning Association*, *76*(1), 42–58. https://doi.org/10.1080/01944360903418290

Jiao, J., & Dillivan, M. (2013). Transit deserts: The gap between demand and supply. *Journal of Public Transportation*, *16*(3), 23–39. https://doi.org/10.5038/2375-0901.16.3.2

Kain, J. F. (1968). Housing segregation, negro employment, and metropolitan decentralization. *The Quarterly Journal of Economics*, *82*(2), 175–197. https://doi.org/10.2307/1885893

Litman, T. (2002). Evaluating transportation equity. *World Transport Policy and Practice*, *8*(2), 50–65.

U.S. Environmental Protection Agency. (2021). *Smart location database: Version 3.0 technical documentation and dataset*. https://www.epa.gov/smartgrowth/smart-location-mapping